Algebra II with Trigonometry Chapter 11.2

Olympiad Problems (GPT-5.4)

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Problems

1. An ellipse is centered at the origin with foci on the x-axis. Its eccentricity is $\frac{3}{5}$, and it passes through $(4,3)$. Find its equation.

2. Two ellipses are given by

$$\frac{x^2}{25}+\frac{y^2}{9}=1 \qquad \text{and} \qquad \frac{x^2}{16}+\frac{y^2}{16-k}=1.$$

For what value of k do the ellipses intersect in exactly four points forming a square?

3. An ellipse centered at the origin has vertices $(\pm 10,0)$. A point P on the ellipse is such that its distances to the two foci are 6 and 14. Find the coordinates of P.

4. A vertical ellipse centered at the origin passes through $(3,4)$ and has eccentricity $\frac{4}{5}$. Find its equation.

5. Let

$$E_1: \ \frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \qquad E_2: \ \frac{x^2}{b^2}+\frac{y^2}{a^2}=1 \quad (a>b>0).$$

Find all intersection points of E_1 and E_2, and show that they are independent of the eccentricity.

6. An ellipse centered at the origin has major axis on the x-axis. Its minor axis has length 12, and one focus is at $(8, 0)$. Without first finding the full equation, determine the sum of the distances from the point $(6, 0)$ on the major axis to the two vertices of the ellipse.